INCH-POUND

MIL-P-82809(OS)
29 December 1989
SUPERSEDING
WS 20630A
16 June 1986

MILITARY SPECIFICATION

POLYMER, LIQUID, HYDROXYL TERMINATED POLYBUTADIENE, WITH ANTIOXIDANT

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 **Scope.** This specification covers the requirements for one type of hydroxyl terminated polybutadiene liquid polymer blended with 2,2'-methylene bis (4-methyl, 6-tert-butylphenol) antioxidant compound, referred to herein as the polymer (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 **Government documents.**

2.1.1 **Specifications, standards and handbooks.** The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6810

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

FED-STD-313 Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402-0001.)

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-5094.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606-4677.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **First article.** When specified (see 6.2), a sample of the polymer shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 **Material.** The polymer shall consist of hydroxyl terminated liquid polymer with 1.0 percent antioxidant compound.

3.3 **Physical and chemical properties.** Physical and chemical properties of the polymer shall conform to the requirements specified in table I.

TABLE I. Physical and chemical properties.

Property	Limits	
	Minimum	Maximum
Hydroxyl value, meq/gram	0.70	0.80
Peroxide content, hydrogen peroxide (ppm)	---	500
Moisture content, percent	---	0.10
Viscosity, poises at 30 degrees Celsius(^{0}C)	40	60
Specific gravity, 25^{0}C /25^{0}C	0.870	---
Iron content, ppm Fe	---	100
Antioxidant content, percent	0.70	1.30

3.4 **Shelf life.** The polymer specified herein shall meet the requirements of this specification for a period of 12 months from the date of the quality conformance inspection (see 4.4) when packaged as specified in section 5 and stored at 80 $\pm$ 20^{0}F.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

3.4.1 **Allowable temperature excursions.** A cumulative total of 30 days maximum outside the 80 ± 20^0F shelf life temperature shall be allowed. Polymer that has exceeded 30 days cumulative total outside these temperature limits may have the shelf life extended as specified in 3.4.2.

3.4.2 **Shelf life extension.** The shelf life of an individual lot may be extended for additional 12-month periods provided the polymer, upon retest, conforms to the hydroxyl number, moisture content, and viscosity requirements.

3.5 **Safety.**

3.5.1 **Material Safety Data Sheet (MSDS).** The contractor shall prepare and submit a MSDS in accordance with FED-STD-313 as specified in the contract (see 6.5).

3.5.2 **Toxic products and formulations.** The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.1.2).

3.6 **Workmanship.** Workmanship shall be such that the polymer is uniform in appearance and free from visible contamination which is foreign to the process of manufacture. (Trace amounts of finely divided particulate matter are common to this material and shall not be cause for rejection).

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:41Z
Check the source to verify that this is the currentversion before use.

4.1.2 **Toxicological product formulations.** The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

4.2 **Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 **First article inspection.** First article inspection shall consist of the quality conformance inspections (see 4.4). Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

4.4 **Quality conformance inspection.** Quality conformance inspection shall consist of the examination and tests specified in table II.

TABLE II. First article and quality conformance inspections.

Tests and Inspections	Requirement	Test Method
Visual examination	3.6 and section 5	4.5.1
Hydroxyl value test	3.3	4.5.2
Peroxide content test	3.3	4.5.3
Moisture content test	3.3	4.5.4
Viscosity test	3.3	4.5.5
Specific gravity test	3.3	4.5.6
Iron content test	3.3	4.5.7
Antioxidant content test	3.3	4.5.8

4.4.1 **Sampling for inspection.**

4.4.1.1 **Lot.** For inspection purposes, a lot shall consist of all the polymer subject to inspection at one time from one of the following:

a. A single batch (see 6.4.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

b. A continuous process (see 6.4.2).

c. A uniform blend of batches or a uniform blend of portions from a continuous process.

4.4.1.2 **Selection of sample containers.** Sample containers of the polymer shall be selected at random from each lot in accordance with table III for the inspections specified in 4.5.1 through 4.5.8. Sample material shall be selected as specified in 4.4.1.3.

TABLE III. Sampling for inspection.

Number of containers in the lot	Number of sample containers to be selected for inspection
1 - 3	All
4 - 40	3
41 - 65	4
66 - 110	5
111 - 180	7
181 - 300	10
301 - 500	15

4.4.1.3 **Sample material.** Sufficient material shall be selected at random from each sample container to perform all of the inspections specified in 4.5.1 through 4.5.8.

4.4.1.4 **Noncompliance of samples.** Failure of one or more of the sample containers of material to meet all of the specified requirements shall be cause for rejection of the lot.

4.4.1.5 **Sample preparation at the procuring activity.** The single samples from a lot shall be combined and blended to form three composite samples. An individual single sample shall not be split between composites, except where a lot consists of fewer than three containers. A single determination of each quality characteristic listed in table I shall be performed on each of the three composite samples independently. When a lot consists of fewer than three containers, three samples shall be taken from each container. The sample portions thus obtained shall be grouped and blended to form three composite samples. The results of each test shall conform to the requirements of table I. Failure of any composite sample to meet any requirement shall be cause for rejection of the lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

4.5 **Methods of inspection.** The following inspection methods shall be used. Unless otherwise specified in the method, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all tests defined in this specification. Solvents and indicators may be commercial nonreagent grade materials.

4.5.1 **Visual examination.** Each sample of material shall be visually examined for conformance to 3.6 and section 5.

4.5.2 **Hydroxyl value test.** Hydroxyl value shall be determined in accordance with the following.

a. Introduce 25.0 milliliters (ml) of phthalic anhydride solution (prepared by dissolving 112 grams (g) of phthalic anhydride in 800 ml of pyridine) into a 200-ml pressure bottle, purge with nitrogen for 2.0 ± 0.4 minutes, then stopper the bottle.

b. Weigh a 4- to 5-g sample (to the nearest 0.1 mg) by difference into the pressure bottle (reseal the bottle as quickly as possible).

c. Warm the bottle and contents under hot water, then immerse the bottle to the liquid level in a steam or oil bath in a ventilated hood.

d. Maintain the temperature at 95 to 98°C for 2 hours minimum, then cool to room temperature.

e. Add 60 ml of tetrahydrofuran.

f. Add 40.0 ml of 0.6N standardized sodium hydroxide, 5 drops of phenolphthalein indicator solution (1g/100 ml pyridine), and titrate to a pink endpoint which persists for at least 15 seconds with 0.6N sodium hydroxide.

g. Determine one or two reagent blanks.

h. Calculate hydroxyl value as:

$$\text{Hydroxyl value (meq/g)} = \frac{(B - A)\,N}{W}$$

Where: B = volume sodium hydroxide to titrate sample, ml

A = volume sodium hydroxide to titrate blank, ml

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

N = normality of sodium hydroxide

W = sample weight, g

i. Record hydroxyl value to the nearest 0.01 meq/g.

4.5.3 **Peroxide content test.** Peroxide content shall be determined in accordance with the following:

a. Weigh, to the nearest 0.1 mg, a 2.5 ± 0.2 g sample of polymer into a flask.

b. Dissolve the polymer in 25 ± 5 ml of carbon tetrachloride.

c. Add small pieces of dry ice to the flask and loosely stopper the flask, allowing the carbon dioxide to purge the flask.

d. Add 20 ± 4 ml of acetic acid solution (prepared by diluting 4.0 ± 0.8 ml of concentrated HCl to 1 liter with glacial acetic acid and purging for 15 ± 3 minutes with nitrogen) and more small pieces of dry ice, and loosely stopper the flask.

e. When the dry ice is gone, warm the flask to room temperature using tap water.

f. Add 2.0 ± 0.4 ml of potassium iodide solution (prepared by dissolving 60.0 ± 0.2 g of potassium iodide in 50 ml distilled water which has been purged with nitrogen for 5 ± 1 minutes), stopper the flask, and mix well.

g. Allow the reaction to proceed for 5 minutes ± 10 seconds, add 100 ml of distilled water, and mix thoroughly for 1.0 ± 0.2 minute.

h. Titrate the sample with standardized 0.01 N sodium thiosulfate solution prepared by diluting 25 ml of 0.1 N sodium thiosulfate (prepared by dissolving 25 g of sodium thiosulfate pentahydrate in 1 liter of freshly boiled, cooled, distilled water to which 0.1 g of sodium carbonate has been added) to 250 ml.

NOTE: Standardize 0.1 N thiosulfate solution with potassium dichromate using 1 percent starch solution as indicator, and calculate as:

$$N = \frac{W}{V\ 0.04904}$$

Where: W = weight of potassium dichromate (approximately 0.2 g)

V = volume of 0.1 N sodium thiosulfate solution, ml

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

0.04904 = gravimetric factor for normality of thiosulfate solution

The normality of the 0.01 N thiosulfate solution is 0.1 of N above.

i. Titrate the sample to a light yellow color, add 5 ± 1 ml of 1 percent starch solution (prepared by dissolving 1 g of soluble starch powder plus a few milligrams of mercuric iodide in 100 ml of boiling water), and continue the titration to the disappearance of the blue color.

NOTE: If the peroxide concentration is small, the starch may be added at the beginning of the titration.

j. Calculate peroxide content as:

$$\text{Peroxide content (ppm)} = \frac{V\ N\ 17{,}000}{W}$$

Where: V = volume of 0.01 N sodium thiosulfate (ml)

N = normality of 0.01 N sodium thiosulfate titrant

W = sample weight (g)

17,000 = gravimetric factor for percent peroxide in this reaction

k. Record peroxide content to the nearest ppm.

4.5.4 **Moisture content test.** Moisture content shall be determined in accordance with the following:

a. Add 10 ± 2 g of sample (weighed to the nearest 0.1 mg) to 100 ± 20 ml of neutralized 1:1 pyridine-chloroform carrier solution or 2 to 4 g of sample to 90 ± 18 ml of 90:10 chloroform-methanol (0.5 percent water) carrier solution.

b. Titrate the moisture using a Karl Fischer (KF) titration assembly with standardized KF reagent (water equivalence of 1.0 to 1.5 mg water per ml of reagent) (or 0.7 ± 0.1 mg water per ml of reagent).

c. Consider the determination complete when an endpoint is reached and remains stable for 30 seconds.

d. Calculate the percent moisture as:

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:41Z
Check the source to verify that this is the currentversion before use.

$$\text{Percent moisture} = \frac{VE}{10W}$$

Where: E = water equivalent of KF reagent, mg water per ml

V = volume of dilute KF reagent, ml

W = sample weight, g

10 = conversion factor to convert mg/ml to percent

e. Record moisture content to the nearest 0.01 percent.

f. The water equivalence of KF reagent shall be determined by adding 2 to 10 mg of water (commercial standard by volume or accurately weighed water by difference from a syringe have been found acceptable) to a neutralized carrier (see a above). Titrate the water as in b above.

g. Calculate water equivalence as:

$$\text{Water equivalent of KF (E)} = \frac{A}{V}$$

Where: A = water added (mg)

V = volume of KF reagent (ml)

Duplicate determinations should agree within 0.04 mg water per ml KF reagent.

4.5.5 **Viscosity test.** Viscosity shall be determined in accordance with the following:

a. Place a minimum of 1 pint of sample in a temperature controlled bath at 30.0 $\pm$ 0.1^{0}C for 4 hours minimum to condition to temperature.

b. Insert a No. 4 Brookfield viscosity spindle into the sample, immersing the spindle to the depth indicated by the index on the spindle.

c. Attach the spindle to a Brookfield RVF viscometer set for 20 revolutions per minute (rpm) rotation.

d. Measure the viscosity during a minimum of five revolutions.

e. Record viscometer dial reading in poises.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

4.5.6 **Specific gravity test.** Specific gravity shall be determined in accordance with the following:

a. Weigh accurately a standardized, clean, dry pycnometer.

b. Slightly overfill pycnometer with sample, keeping trapped air at a minimum, and insert the top loosely.

c. Cool the assembly to 23^0C or lower, then push the top down tightly with a twisting motion.

d. Immerse filled pycnometer in a water-filled constant temperature bath controlled to 25.0 ± 0.1^0C, and allow the sample to come to temperature equilibrium.

e. Wipe off excess sample, dry thoroughly, and weigh.

f. Calculate specific gravity as follows:

$$\text{Specific gravity, } 25^0\text{C}/25^0\text{C} = \frac{A - B}{0.997V}$$

Where: A = weight of filled pycnometer, g

B = weight of empty pycnometer, g

V = volume of calibrated pycnometer, ml

0.997 = density of water at 25^0C.

g. Record specific gravity to the nearest 0.001 unit.

4.5.7 **Iron content test.** Iron content shall be determined in accordance with either 4.5.7 or 4.5.7.1.

a. Weigh 10 ± 2 g of sample to the nearest 0.1 mg into a separatory funnel containing 40 ± 8 ml of carbon tetrachloride.

b. Add 20 ml of wash solution (prepared by mixing 45 parts distilled water, 45 parts denatured ethanol, and 6 parts concentrated hydrochloric acid). Shake the separatory funnel to ensure intimate contact between phases. Allow the phases to separate, and collect the aqueous washings in a beaker.

c. Repeat b for a total of four washings.

d. Evaporate the washings to less than 15 ml on a hotplate, cool, and carefully add 3.0 ± 0.6 ml of concentrated sulfuric acid.

e. Heat the beaker and concentrated washings to drive off the water.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

f. Cover the beaker with a watch glass, and digest the sample with small amounts of 50 percent hydrogen peroxide.

g. Cool the digested sample, dilute with 40 ml of distilled water, and adjust the pH to 6 to 7 with ammonium hydroxide. Dissolve 2 to 4 g of ascorbic acid in the sample, and transfer the solution to a 100-ml volumetric flask. Pipette 10.0 ml 2,2'-bipyridine solution (prepared by dissolving 1 g of 2,2'-bipyridine in 100 ml of absolute ethanol and diluting to 1 liter with distilled water) into the flask, and dilute to volume with distilled water.

h. Mix thoroughly, and allow the color to develop for at least 60 minutes.

i. Measure the absorbance of the iron-bipyridine complex on a spectrophotometer at 510 nanometers (nm).

j. Prepare a standard curve covering the anticipated iron concentration range with a minimum of three points, using pure iron wire as the source of iron.

k. Calculate the iron concentration in the sample by comparison with the standard curve, and record the iron in parts per million to the nearest ppm.

4.5.7.1 **Alternative iron content test.**

a. Prepare iron standards by diluting commercial iron-containing standard (5000 ppm iron has been found satisfactory for this purpose) with low iron polymer. (Standards in the range from 0.001 to 0.020 percent iron have been found satisfactory for this analysis.)

b. Analyze a blank and each standard using x-ray fluorescence spectrometry, following the equipment manufacturer's operating procedures and instrument settings at the background angle and at the FeK (alpha) angle.

c. Plot the net counts versus weight-percent iron for each standard.

d. Analyze samples using the same technique as for the standard.

e. Read the weight percent iron in the samples from the standard curve.

f. Record iron content to the nearest 0.001 percent.

4.5.8 **Antioxidant content test.** The antioxidant content shall be determined as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

a. Weigh a 0.20 ± 0.04-g sample of polymer to the nearest mg into a 25-ml volumetric flask, dissolve, and dilute to volume with p-dioxane.

b. Pipette a 5.00-ml aliquot of the sample solution to each of two 50-ml volumetric flasks labeled "A" and "B". To flask "A" add 5.00 ml of 0.018 N TBAH. Prepare as TBAH by diluting 15 ml of 25 percent methanolic tetrabutyl ammonium hydroxide to 1 liter with methanol. Determine the normality of the TBAH by titration with 0.01 N HCl (0.3 percent alcoholic thymol blue indicator), and adjust the concentration if necessary. Dilute the TBAH to volume with p-dioxane. To flask "B" add 5.00 ml of methanol, and dilute to volume with p-dioxane.

c. Using solution "B" as a blank, measure the absorption of solution "A" at 315 nm using 1-cm silica cells and a bandwidth of 1.0 to 1.5 nm in an ultraviolet spectrophotometer. Read the antioxidant content from the standard curve.

d. Prepare a standard stock solution by weighing, to the nearest 0.1 mg, 0.10 to 0.12 g of antioxidant (purified by crystallizing commercially available 2,2' methylene bis (4 methyl, 6-tert-butylphenol) from ligroin or ether to yield white needles melting at 130 to 131^{0}C) into a 250-ml volumetric flask and diluting to volume with p-dioxane.

e. Pipette 1, 2, 3, 4, (5 optional) ml aliquots respectively of the standard stock solution into 50-ml volumetric flasks. Pipette 5.00 ml 0.018 N TBAH in methanol into each flask, and mix and dilute to volume with p-dioxane.

f. Prepare a blank by diluting 5.00 ml of 0.018 N TBAH in methanol to 50 ml with p-dioxane.

g. Measure the absorbance of each standard solution against the blank at 315 nm. Plot the absorbance versus antioxidant using linear graph paper.

h. Calculate the standard antioxidant concentration as follows:

(mg antioxidant/50-ml) = 4 x A x B

Where: A = antioxidant

B = aliquot of stock solution

4 = concentration/unit conversion factor

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

$$\text{Antioxidant concentration (percent)} = \frac{0.5\ C}{W}$$

Where: C = mg antioxidant/50-ml from plotted curve

W = sample weight (g)

0.5 = factor to account for dilution, conversion from milligrams to grams, and percentage

i. Record the antioxidant content to the nearest 0.01 percent.

4.6 **Inspection of packaging.** The polymer containers and container markings shall be visually examined to verify compliance with section 5.

5. PACKAGING

5.1 **Packaging.** Unless otherwise specified in the contract or order, (see 6.2), packaging of the polymer shall be in level C as specified herein.

5.1.1 **Level C.** The polymer shall be packaged to afford adequate protection against loss, contamination, deterioration and damage during shipment from the supply source to the first receiving activity and during storage under the shelf life period and conditions specified in 3.4. Containers in the same shipment shall be of the same size. The packaging shall conform to UFC 6000, National Motor Freight Classification, 49 CFR 171-178, or to other carrier rules and regulations as applicable to the mode of transportation.

5.2 **Marking.**

5.2.1 **Standard marking.** In addition to any special marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.2 **Special marking.** In addition to the marking requirements of 5.2.1, each container marking shall include the following information:

a. Title, number and date of this specification.

b. Manufacturer's name and address.

c. Material trade name and identification.

d. Net weight of contents.

e. Lot number and date of quality conformance inspection (see 3.4).

f. Shelf life and inspection/test date for extension (see 3.4). In addition, the marking shall include the following:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

"SHELF LIFE EXTENSION

THE SHELF LIFE OF AN INDIVIDUAL LOT MAY BE EXTENDED FOR AN ADDITIONAL 12 MONTHS PROVIDED THE POLYMER, UPON RETEST, CONFORMS TO THE HYDROXYL NUMBER, MOISTURE CONTENT, AND VISCOSITY REQUIREMENTS OF MIL-P-82809".

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The liquid polymer specified herein is intended for use in the manufacture of the Dual Thrust Rocket Motor, Mark 104, of the Standard Missile.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1, 2.1.2 and 2.2).

c. Whether first article inspection is required (see 3.1 and 6.3).

d. Material Safety Data Sheets (see 3.5.1 and 6.5).

e. Special marking, if other than as specified (see 5.2.2).

f. Safety precautions required (see 6.6).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors on the material to be inspected. The material should be a first article sample and should consist of a quantity of the polymer sufficient to perform all of the inspections specified in 4.3. The first article sample should be packaged and marked as specified in section 5. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definitions.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:41Z
Check the source to verify that this is the current version before use.

6.4.1 **Batch.** A batch is that amount of polymer made as one unit in the manufacturing process.

6.4.2 **Continuous process.** A continuous process is an uninterrupted process of manufacture from which, in one unbroken flow, the material is accumulated for a lot.

6.5 **Material Safety Data Sheet (MSDS).** MSDS requirements are applicable to this specification and should be specified in the contract as required by the Federal Acquisition Regulation (FAR), Part 23, Subpart 23.3. Contracting officers will identify those activities requiring copies of completed MSDSs prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.6 **Safety precautions.** The safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract as required by the DOD FAR Supplement 28.7102.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP5) should be made applicable.

6.7 **Possible material.** The following material has been found to meet the requirements of this specification. This is only given for information and is not restrictive. Poly BD R-45M A0-1 percent, manufactured by ARCO Chemical Company, Philadelphia, PA, has been found to meet the requirements of this specification.

6.8 **Subject term (key word) listing.**

Missile, Standard
Rocket Motor, Dual Thrust, Mark 104

Preparing Activity:
NAVY-OS
(Project 6810-N053)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-P-82809	2. DOCUMENT DATE (YYMMDD) 29 December 1989

3. DOCUMENT TITLE
POLYMER, LIQUID, HYDROXYL TERMINATED POLYBUTADIENE, WITH ANTIOXIDANT

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-26 04:41Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.